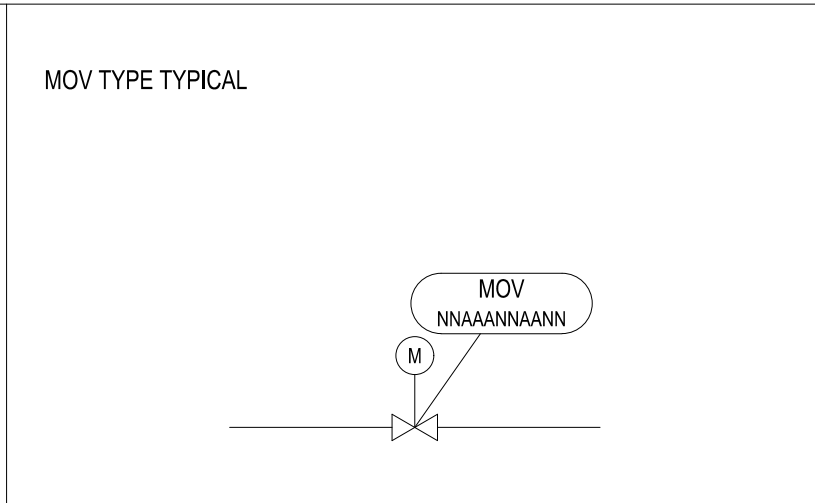
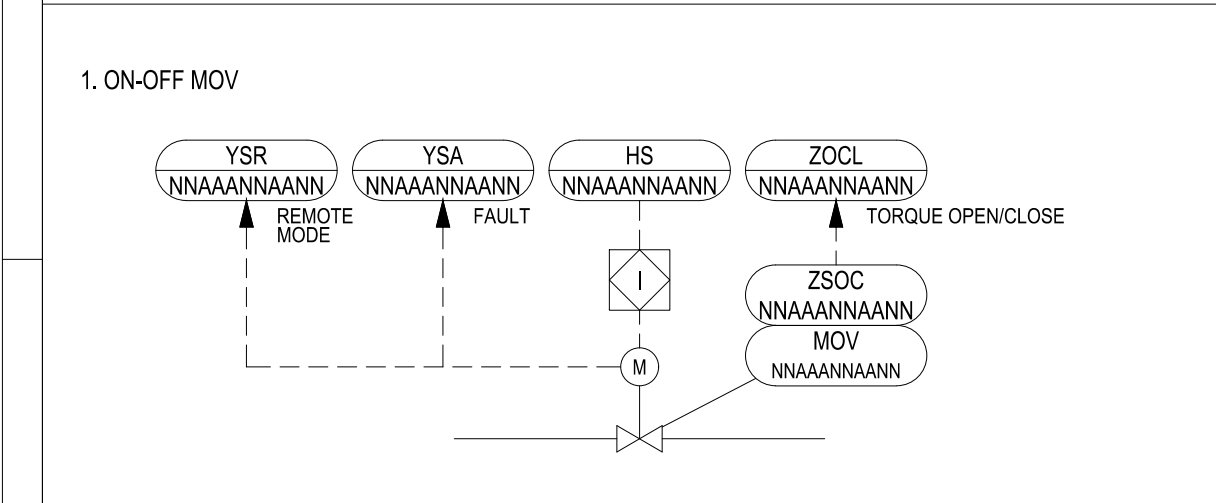


8		7		6		5		4		3		2		1					
FIRST LETTER		SUCCEEDING LETTERS													SELF-ACTUATED DEVICES-PRESSURE		GENERAL NOTES : 1. THIS DRAWING IS PREPARED FOR THE PROPOSAL PURPOSE ONLY AND CAN BE CHANGED ACCORDINGLY IN THE DETAIL DESIGN PHASE.		
		SENSING DEVICE		DISPLAY DEVICE			CONTROL DEVICE				MISCELLANEOUS DEVICE								
SYMBOL	MEASURED OR INITIATING VARIABLE	PRIMARY ELEMENT	TRANSMITTER	INDICATOR	RECORDER	INTEGRATING INDICATOR	ARARM ANNUNCIATOR			CONTROL STATION MAN/AUTO	CONTROLLER	CONTROL VALVE	OTHER FINAL CONTROL ELEMENT	SWITCH	LOCAL OBSERVATION GLASS	SOLENOID RELAY OR POSITIONER	TEST POINT	USER'S CHOICE	UNCLASSIFIED
()	TYPICAL SYMBOL	()E	()T	()I	()R	()QI	()AL	()AH	()AHL	()K	()C	()CV	()Z	()S	()G	()Y	()P	()N	()X
A	ANALYSIS	A E	A T	A I	A R		AAL	AAH	AAHL	A K	A C		A Z	A S		A Y	A P	A N	A X
B	BURNER FLAME	B E	B T	B I	B R		BAL				B C			B S		B Y	B P	B N	B X
C	CONDUCTIVITY	C E	C T	C I	C R		CAL	CAH	CAHL	C K			C Z	C S		C Y	C P	C N	C X
D	DENSITY (MASS) OR SPECIFIC GRAVITY	D E	D T	D I	D R		DAL	DAH	DAHL	D K	D C		D Z	D S		D Y	D P	D N	D X
E	VOLTAGE (EMF)	E E	E T	E I	E R		EAL	EAH	EAHL	E K	E C		E Z	E S		E Y		E N	E X
F	FLOW	F E	F T	F I	F R	FQI	FAL	FAH	FAHL	F K	F C	FCV	F Z	F S	F G	F Y	F P	F N	F X
G	GAUGING (DIMENSIONAL)	G E	G T	G I	G R		GAL	GAH	GAHL	G K	G C			G S		G Y		G N	
H	HAND (MANUAL INITIATED)	H									H C		H Z	H S		H Y		H N	
I	CURRENT	I E	I T	I I	I R	IQI	IAL	IAH	IAHL	I K			I Z	I S		I Y		I N	I X
J	POWER	J E	J T	J I	J R	JQI	JAL	JAH	JAHL	J K			J Z	J S		J Y		J N	J X
K	TIME	K E	K T	K I	K R	KQI	KAL	KAH	KAHL		K C		K Z	K S		K Y		K N	K X
L	LEVEL	L E	L T	L I	L R		LAL	LAH	LAHL	L K	L C	LCV	L Z	L S	L G	L Y	L P	L N	L X
M	MOISTURE OR HUMIDITY	M E	M T	M I	M R		MAL	MAH	MAHL	M K	M C		M Z	M S		M Y	M P	M N	M X
O	USER'S CHOICE																		
P	PRESSURE OR VACUUM	P E	P T	P I	P R		PAL	PAH	PAHL	P K	P C	PCV	P Z	P S		P Y	P P	P N	P X
PD	PRESSURE DIFFERENTIAL	PDE	PDT	PD I	PDR		PDAL	PDAH	PDAHL	PDK	PDC		PDZ	PDS					
Q	QUANTITY	Q E	Q T	Q I	Q R	QQI	QAL	QAH	Q AHL	Q K	Q C		Q Z	Q S		Q Y		Q N	Q X
S	SPEED OR FREQUENCY	S E	S T	S I	S R	SQI	SAL	SAH	SAHL	S K	S C		S Z	S S		S Y		S N	S X
T	TEMPERATURE	T E	T T	T I	T R		TAL	TAH	TAHL	T K	T C	TCV	T Z	T S		T Y	T P	T N	T X
U	MULTIVARIABLE			U I	U R								U Z	U S		U Y		U N	U X
V	VIBRATION		V T	V I	V R		VAL	VAH	VAHL	V K	V C		V Z	V S		V Y	V P	V N	V X
W	WEIGHT OR FORCE	W E	W T	W I	W R	WQI	WAL	WAH	WAHL	W K	W C		W Z	W S		W Y		W N	W X
X	UNCLASSIFIED																		
Y	EVENT OR STATE	Y E	Y T	Y I	Y R		YAL	YAH					Y Z	Y S		Y Y		Y N	Y X
Z	POSITION	Z E	Z T	Z I	Z R		ZAL	ZAH	Z AHL				ZZ	Z S		Z Y		Z N	Z X
GENERAL INSTRUMENTS				SENSOR-FLOW				SENSOR-FLOW (CONT.)				SENSORS-PRESSURE				VALVES ACTUATORS			
FF INSTRUMENT FOR SINGLE MEASURED VARIABLE AND ANY NUMBER OF FUNCTIONS. INSTRUMENT IS LOCALLY MOUNTED. INSTRUMENT FOR SINGLE MEASURED VARIABLE AND ANY NUMBER OF FUNCTIONS. INSTRUMENT IS LOCALLY MOUNTED. FF=FOUNDATION FIELDBUS INSTRUMENT IS MOUNTED ON A LOCAL PANEL (TYPICAL) INSTRUMENT IS MOUNTED ON A MAIN CONTROL PANEL (TYPICAL) INSTRUMENT FOR PERFORMANCE TEST ONLY INSTRUMENT FOR PERFORMANCE TEST PLUS NORMAL USE INSTRUMENT IS MOUNTED BEHIND A PANEL (TYPICAL)				FE FE</															

	8	7	6
H	VALVE BODY WITH ACTUATOR-TYPICAL FOR MOTOR OPERATED VALVES		



2. INCHING

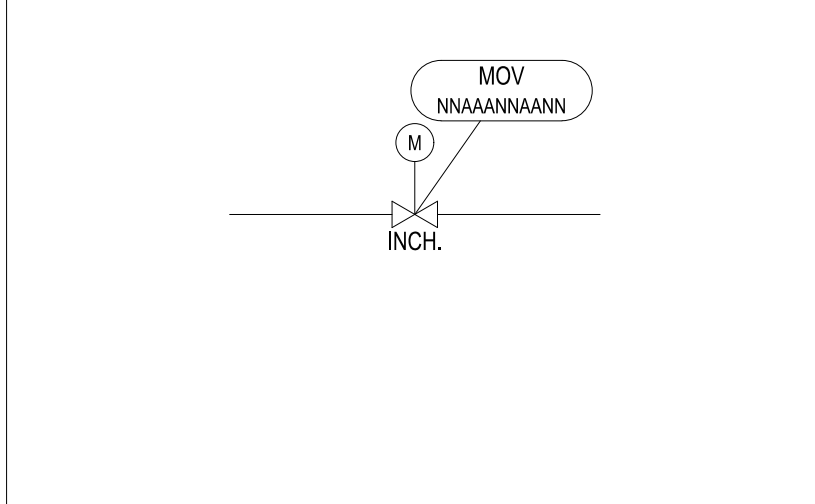
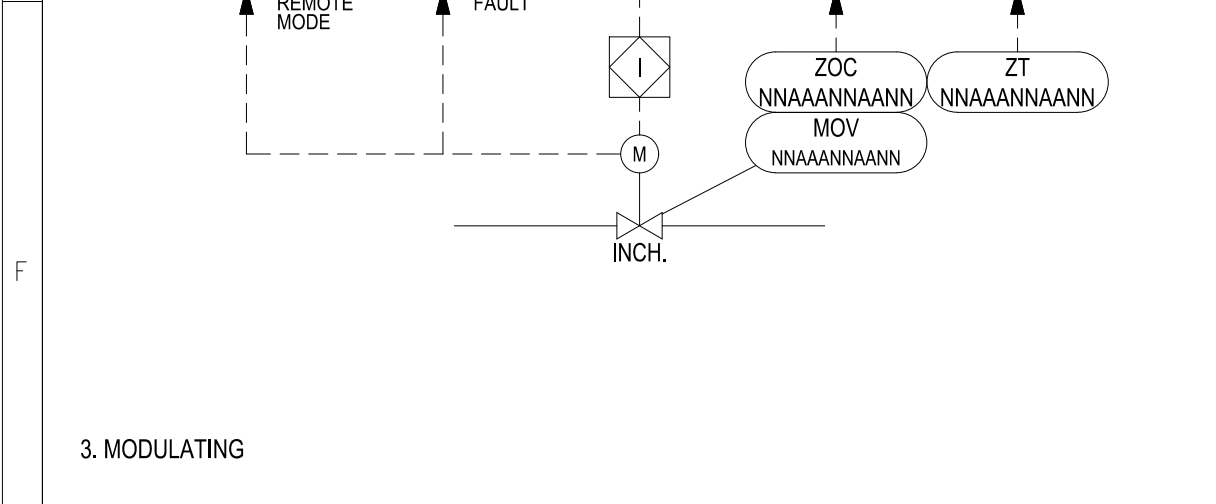
YSR
NNA¹AAANNA¹AA¹NN

YSA
NNA¹AAANNA¹AA¹NN

ZIC
NNA¹AAANNA¹AA¹NN

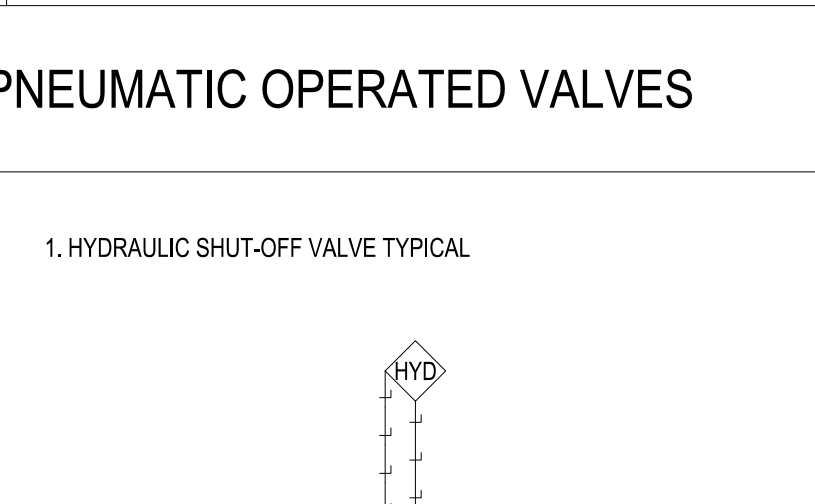
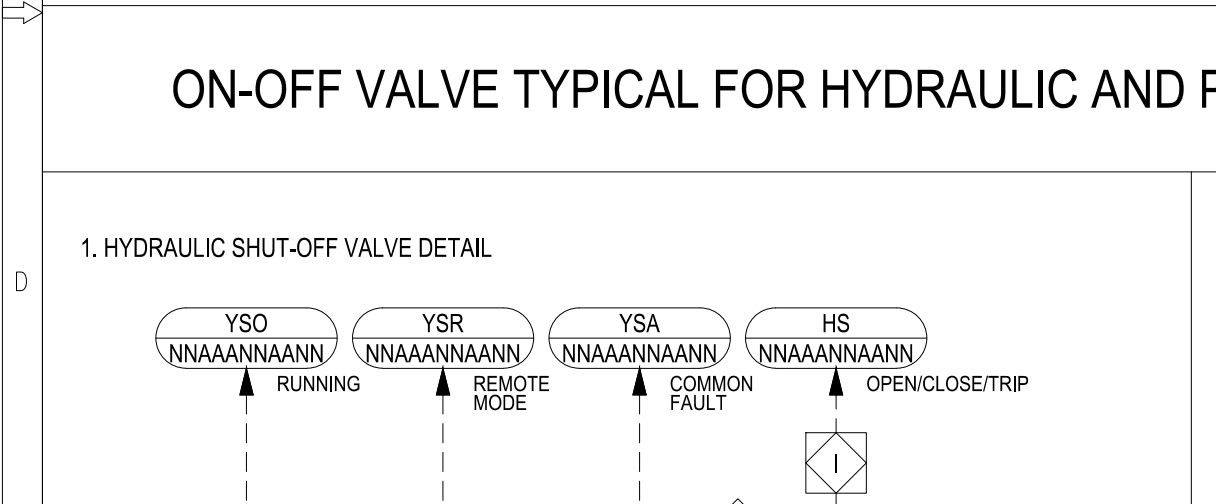
ZOCL
NNA¹AAANNA¹AA¹NN

ZI
NNA¹AAANNA¹AA¹NN

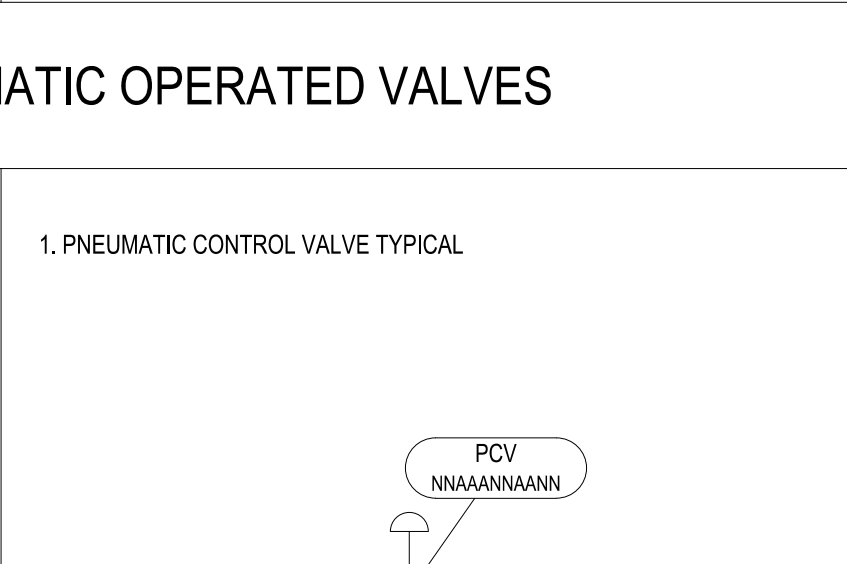
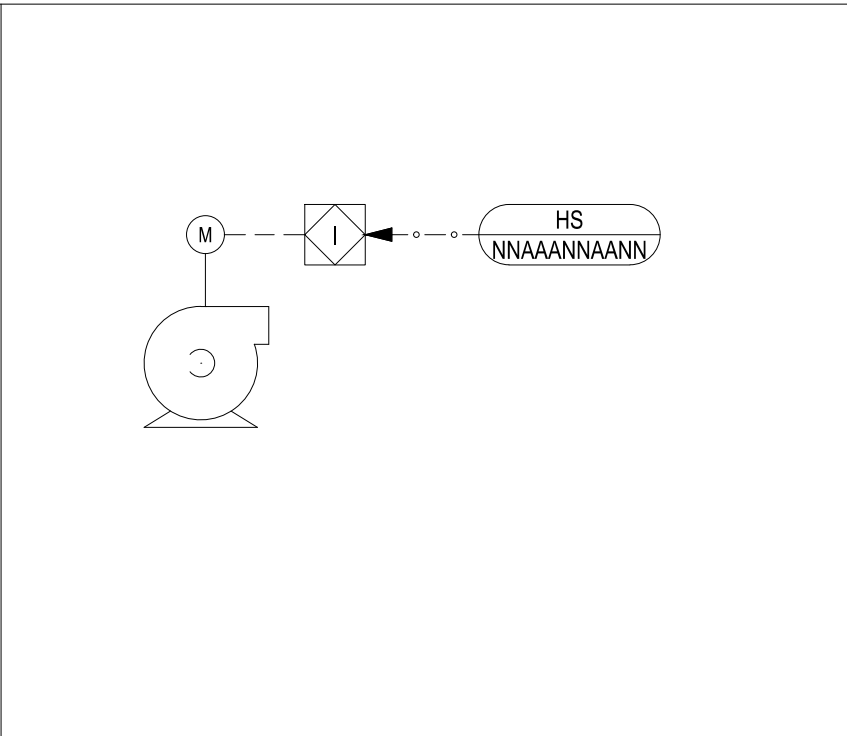


3. MODULATING

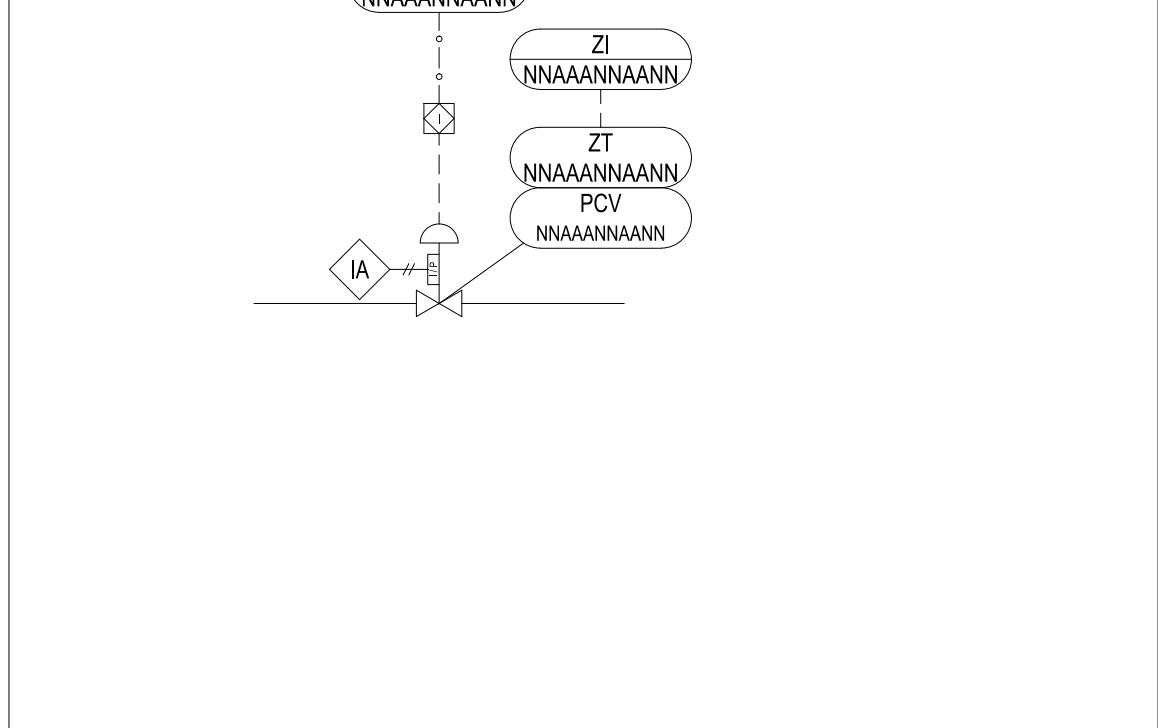
The diagram illustrates the Modulating stage of a 4th-order sigma-delta modulator. It features four input registers: YSR, YSA, ZIC, and ZI, each containing the sequence NNAAAANNAANN. Two output registers, ZT and MOV, also contain NNAAAANNAANN. The inputs are connected to a summing junction (M) via dashed lines labeled 'REMOTE MODE' and 'FAULT'. The summing junction output goes through an integrator (I) and a quantizer (MOD.) to produce the final output.



MOTOR WITHOUT DUTY/STANDBY SELECTION (TYP.1)



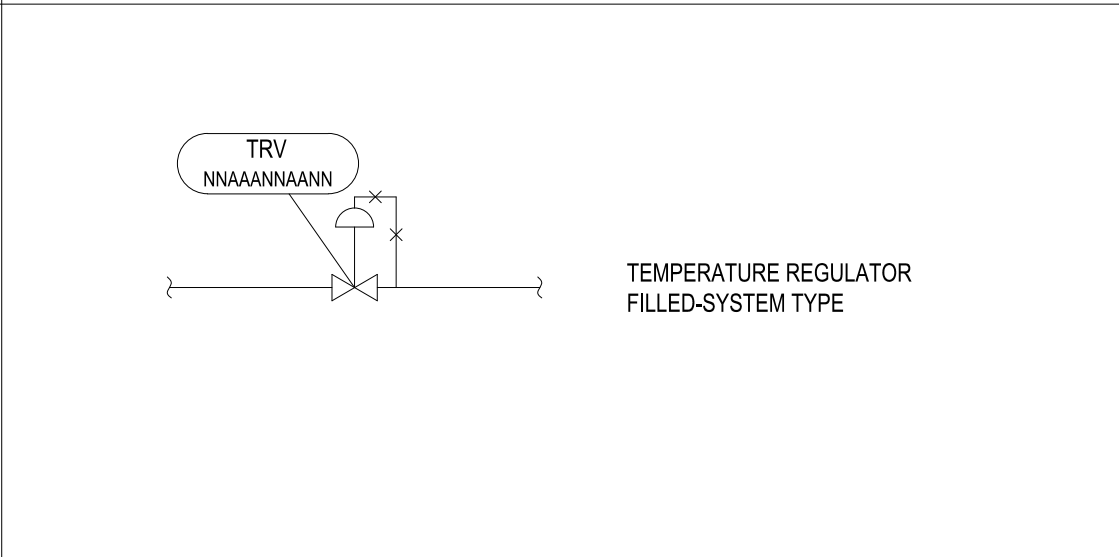
1. PNEUMATIC CONTROL VALVE DETAIL/PNEUMATIC NON-RETURN VALVE DETAIL	1. PNEUMATIC CONTROL VALVE TYPICAL
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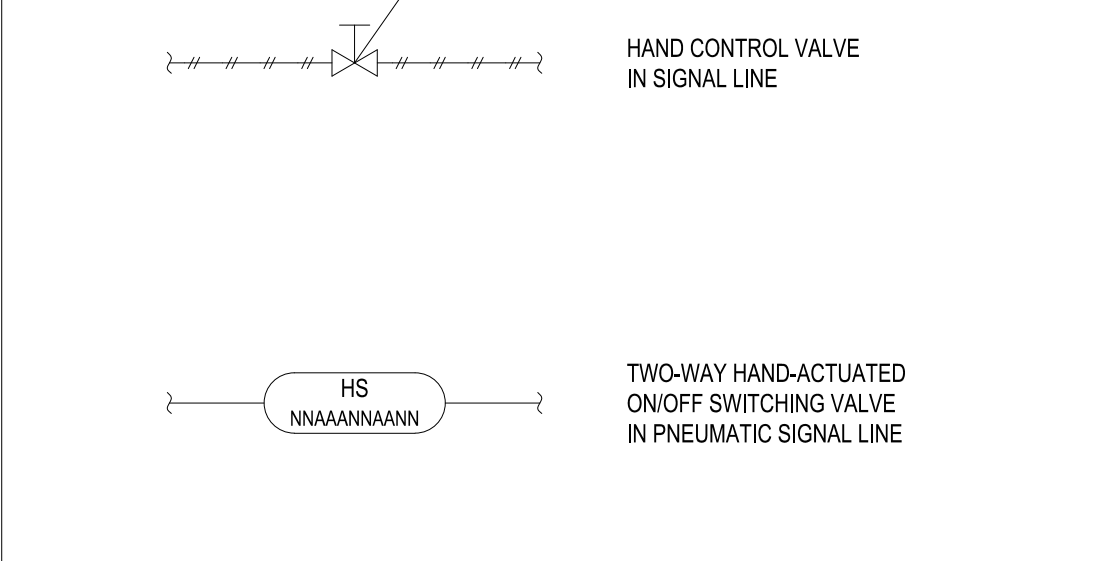
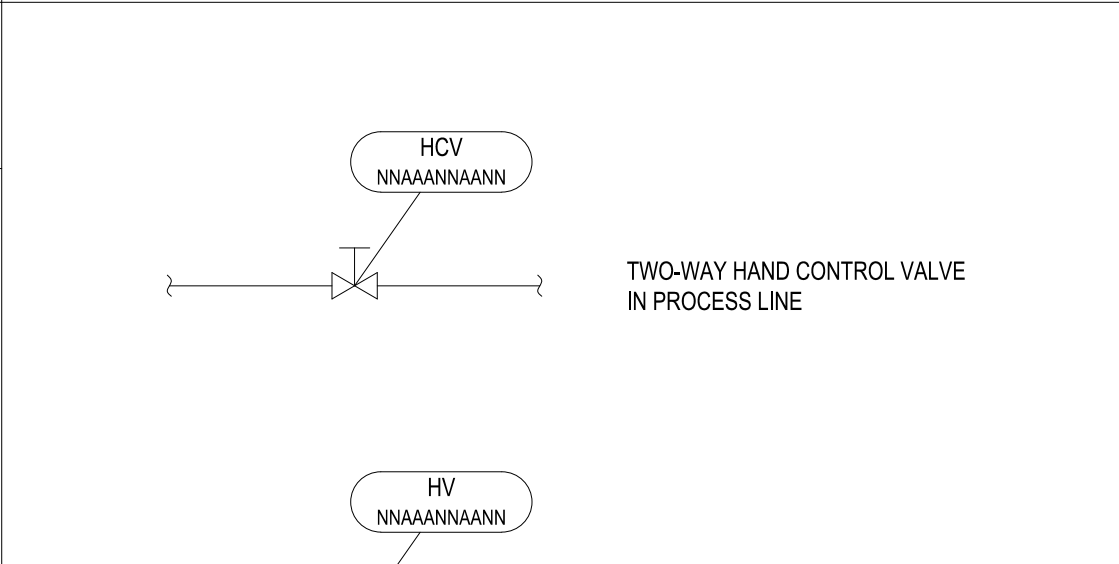
ESDV : EMERGENCY SHUT-DOWN VALVE
SDV : SHUT-DOWN VALVE
SET : SET VALUE
ZSOC: POSITION SWITCH OPEN/CLOSE
ZIOC : POSITION INDICATOR OPEN/CLOSE
ZOCL : POSITION OPEN/CLOSE LAMP

CC : CATIONIC CONDUCTIVITY
SC : SPECIFIC CONDUCTIVITY
XV : SHUT-OFF VALVE
M : ELECTRIC MOTOR
MH : MAN HOLE
SMFCP : SETTLEMENT METERING FLOW COMPUTER PANEL
BMFCP : BACK-UP METERING FLOW COMPUTER PANEL
MECP: MAIN FIRE ALARM CONTROL PANEL
ESEW : EMERGENCY SHOWER & EYE WASHER
DCMS : DISTRIBUTED CONTROL AND MONITORING SYSTEM
BMS : BURNER MANAGEMENT SYSTEM
BPS : BOILER PROTECTION SYSTEM

3	2
SELF-ACTUATED DEVICES-TEMP	



HAND ACTUATED DEVICES



* FURNISHED BY EQUIPMENT SUPPLIER

LOGIC & SEQUENCE CONTROL

OR SEQUENCE CONTROL
NORMALLY USED FOR LOCAL
AUXILIARY CONTROL SYSTEM

NOT NORMALLY ACCESSIBLE TO OPERATOR

- PACKAGED PROGRAMMABLE LOGIC CONTROLLER OR DIGITAL LOGIC CONTROLS INTEGRAL TO THE DISTRIBUTED CONTROL EQUIPMENT

- NORMALLY ACCESSIBLE TO OPERATOR
- PACKAGED PROGRAMMABLE LOGIC CONTROLLER OR DIGITAL LOGIC CONTROLLER INTEGRAL TO THE DISTRIBUTED CONTROL EQUIPMENT

GENERAL NOTES :

1. THIS DRAWING IS PREPARED FOR THE PROPOSAL PURPOSE ONLY AND CAN BE CHANGED ACCORDINGLY IN THE DETAIL DESIGN PHASE.

					R

TENDERER

 **SAMSUNG C&T**
Engineering & Construction Group

PROJECT NAME	AL NOUF1 PROJECT
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DRAWING TITLE

P&ID FOR SYMBOLS & LEGENDS 3

A

PROJECT DWG NO.		D00P-00GEN00-M05-0004	SHEET	REV.
			1/1	A
PROJECT NO.	REF. DRAWING NO.		SCALE	REF. CODE
D00P			NONE.	

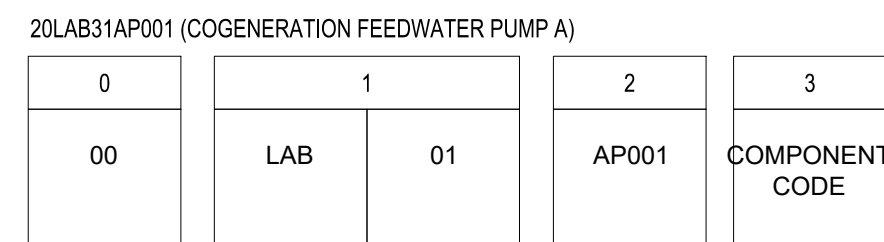
(3) SERIAL NO.

TAG NUMBERING SYSTEM IS BASED ON THE KKS NUMBERING SYSTEM. (DOC. NO.: 19K9-00GEN00-G02-0002)

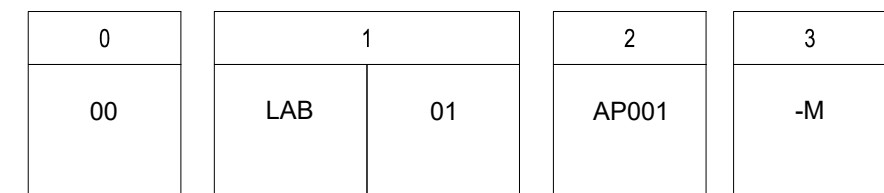


	PLANT UNIT	'UU' CHARACTER DESIGNATION
POWER PLANT	POWER PLANT COMMON	0
	FIRST GROUP COMMON (INCL. FIRST GROUP STG)	10
	FIRST GROUP #1 GTG / HRSG	11
	FIRST GROUP #2 GTG / HRSG	12
	SECOND GROUP COMMON (INCL. SECOND GROUP STG)	20
	SECOND GROUP #1 GTG / HRSG	21
	SECOND GROUP #2 GTG / HRSG	22

EXAMPLE OF APPLICATION:



20LAB11AP001-M (COGENERATION FEEDWATER PUMP A MOTOR)



The diagram illustrates a Piping Design Code (PDC) structure. It starts with a box labeled '11LAB11' and 'BRO01' at the top left, and 'BREAKDOWN LEVEL 0 & 1' at the top right. Below these is a box labeled '250H LAB1' and 'BREAKDOWN LEVEL 2'. Arrows point from these boxes to a list of items: 'PROCESS DESIGN CODE', 'PIPING INSULATION', '- H : INSULATION FOR HEAT CONSERVATION', '- P : INSULATION FOR PERSONAL PROTECTION', '- N : NO INSULATION', and 'PIPING DIAMETER'.

FLOW DIRECTION

11LAB11
BRO01

250H LAB1

BREAKDOWN LEVEL 0 & 1
BREAKDOWN LEVEL 2

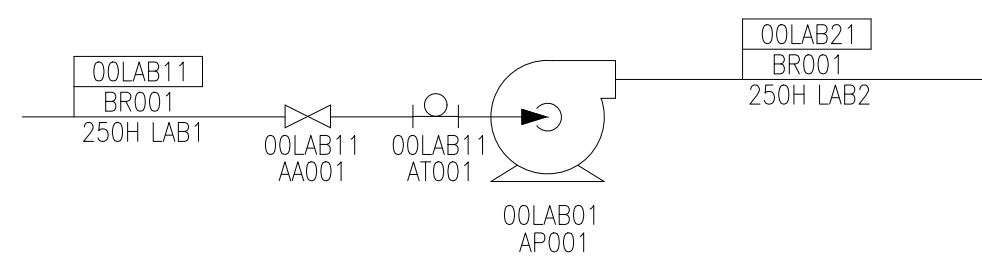
PROCESS DESIGN CODE

PIPING INSULATION

- H : INSULATION FOR HEAT CONSERVATION
- P : INSULATION FOR PERSONAL PROTECTION
- N : NO INSULATION

PIPING DIAMETER

EXAMPLE FOR LINE VALVE, EQUIPMENT AND STRAINER IDENTIFICATION



1. THIS DRAWING IS PREPARED FOR THE PROPOSAL PURPOSE ONLY AND CAN BE CHANGED ACCORDINGLY IN THE DETAIL DESIGN PHASE.

NOTES :

NUMBER RANGE	VALVES	PIPE BR	INSTRUMENTATION CIRCUITS
001-100	-VALVES IN MAIN PIPES -VALVE GENERAL	-MAIN PIPES	-TRANSMITTER
101-190	-CONTROL VALVES / DAMPERS		-SWITCH
191-199	- SAFETY VALVES	-SUCTION AND PRESSURE RELIEF LINES ON SAFETY VALVES	
201-250	-TCE IS USING FOR MOV IN MAIN LINES	-NOT USED	-FOR TESTING EQUIPMENTS
251-299	-SHUT OFF VALVE MAIN LINE		
301-399	-INSTRUMENT ISOLATION	-PRESSURE LINES FOR INSTRUMENTS	-NOT USED
401-499	-VALVES IN DRAINS AND FLUSHING PIPES	-DRAINS, FLUSHING PIPES, PRESSURE SUPPRESSION PIPES	-ACCEPTANCE TESTS -TEST THERMOWELL
501-599	-VALVES FOR VENTING	-VENTING	-LOCAL MEASURING CIRCUITS -GAUGES
601-699	-VALVES FOR SAMPLING AND DOSING	-SAMPLING AND DOSING PIPES	-FLOW ELEMENT
701-799	-VALVES FOR INTERNAL CONTROLS	-FOR INTERNAL CONTROLS	-GROUPED
801-899	1. TCE IS USING FOR MOV IN DRAIN LINES 2. STEAM TRAP	-NOT USED	-GROUPED
901-999	-TCE IS USING FOR MOV IN VENT LINES	-NOT USED	-GROUP MEASURING CIRCUITS (E.G. MULTI POINT SELECTOR SWITCHES, MULTI CHANNEL RECORDERS)

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REV.	DATE	DESCRIPTION	PREPARED	REVIEWED	APPROVED
EMPLOYER					

TENDERER



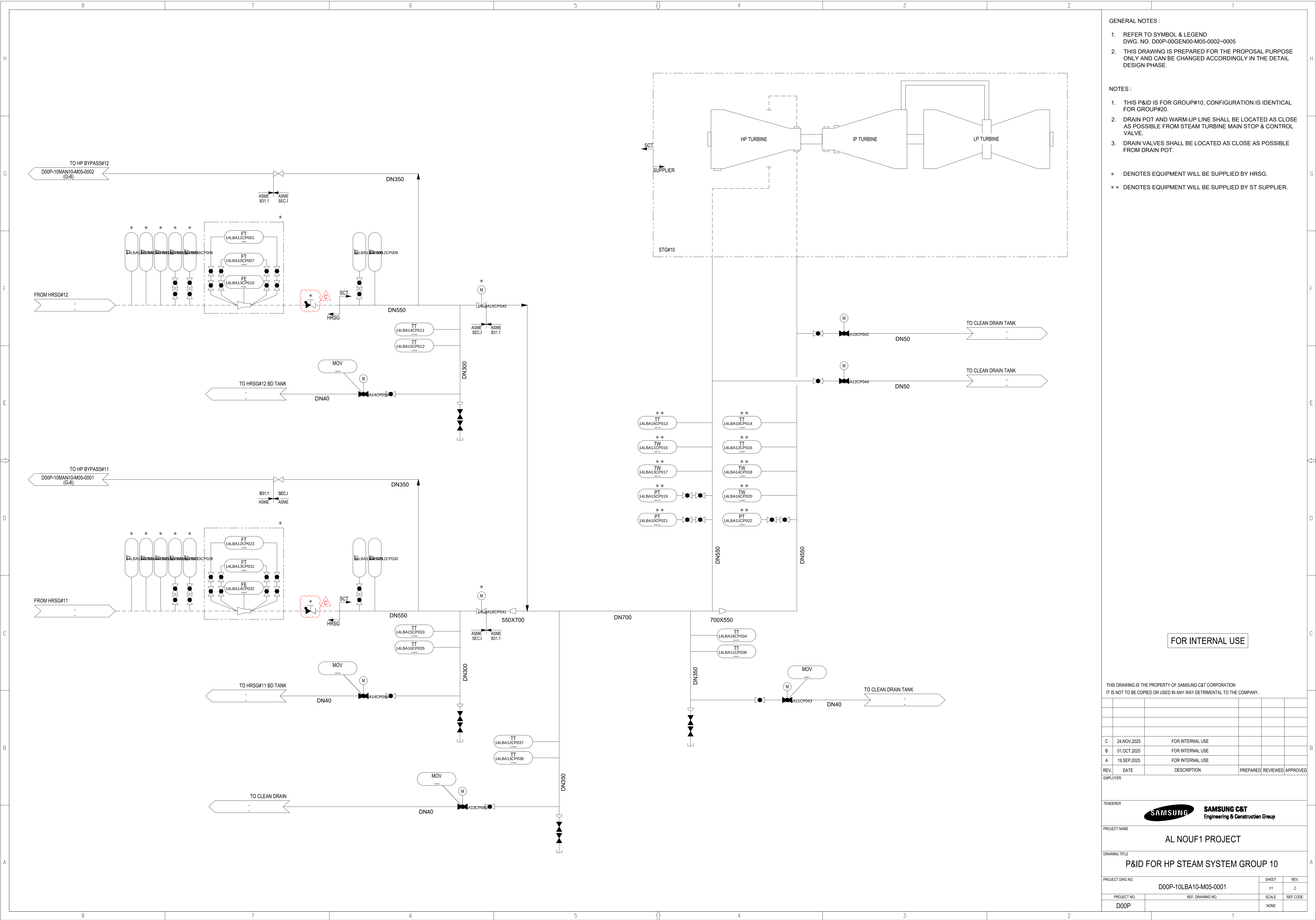
PROJECT NAME

AL NOUF1 PROJECT

DRAWING TITLE

P&ID FOR SYMBOLS & LEGENDS 4

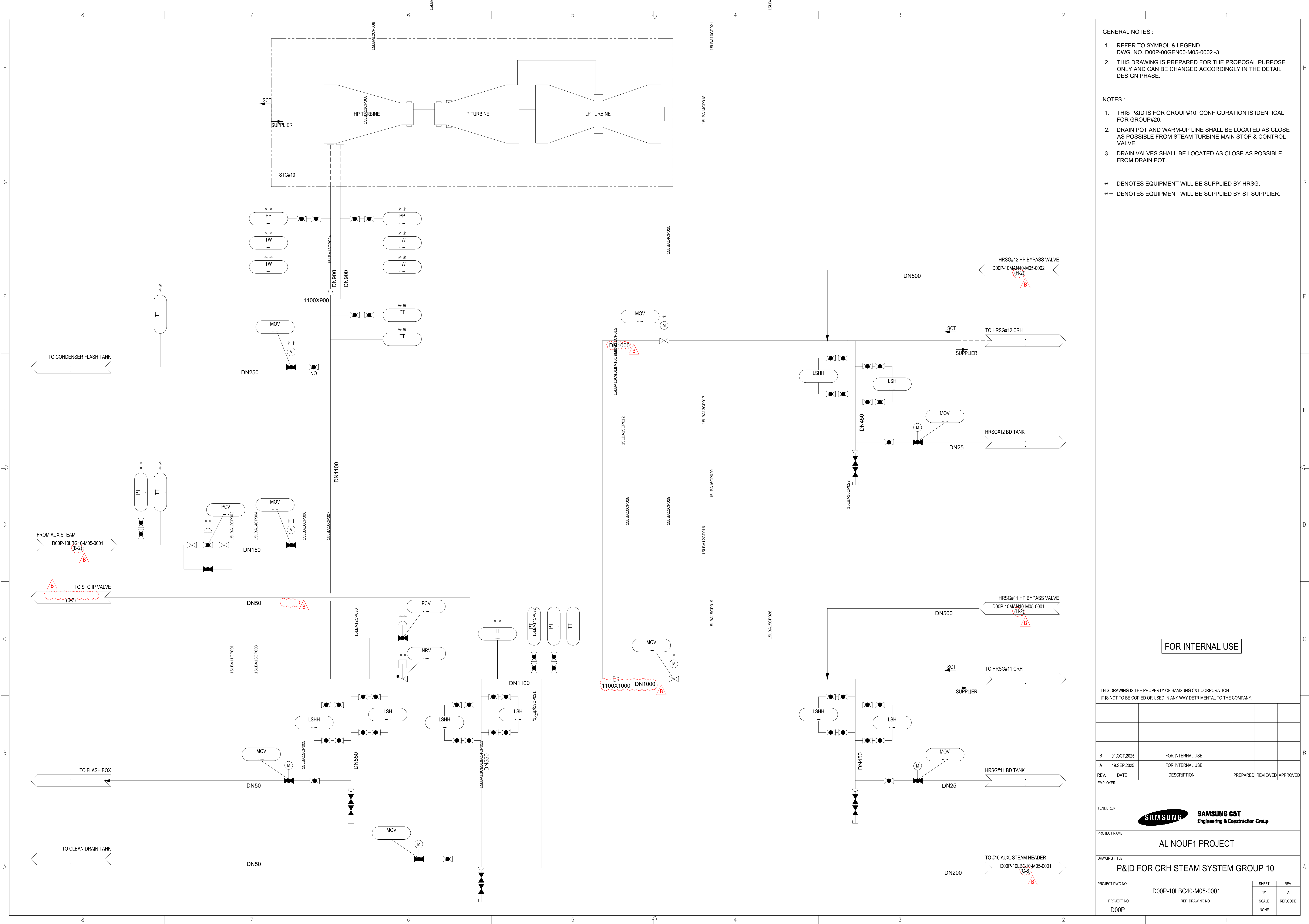
PROJECT DWG NO. D00P-00GEN00-M05-0005		SHEET	REV.
		1/1	A
PROJECT NO.	REF. DRAWING NO.	SCALE	REF. CODE
D00P		NONE	



- GENERAL NOTES :
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DWG. NO. D00P-00GEN00-M05-0002~0005
 - THIS DRAWING IS PREPARED FOR THE PROPOSAL PURPOSE ONLY AND CAN BE CHANGED ACCORDINGLY IN THE DETAIL DESIGN PHASE.
- NOTES :
- THIS P&ID IS FOR GROUP#10, CONFIGURATION IS IDENTICAL FOR GROUP#20.
 - DRAIN POT AND WARM-UP LINE SHALL BE LOCATED AS CLOSE AS POSSIBLE FROM STEAM TURBINE MAIN STOP & CONTROL VALVE.
 - DRAIN VALVES SHALL BE LOCATED AS CLOSE AS POSSIBLE FROM DRAIN POT.
- * DENOTES EQUIPMENT WILL BE SUPPLIED BY HRSG.
** DENOTES EQUIPMENT WILL BE SUPPLIED BY ST SUPPLIER.

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B	01.OCT.2025	FOR INTERNAL USE			
A	19.SEP.2025	FOR INTERNAL USE			
REV.	DATE	DESCRIPTION	PREPARED	REVIEWED	APPROVED
EMPLOYER					
TENDERER					
PROJECT NAME					
DRAWING TITLE					
PROJECT DWG NO.					
PROJECT NO.					
D00P					
D00P-10LBA10-M05-0001				SHEET	REV.
REF. DRAWING NO.				1/1	C
NONE				SCALE	REF. CODE



- GENERAL NOTES :
- REFER TO SYMBOL & LEGEND
DWG. NO. D00P-00GEN00-M05-0002~3
 - THIS DRAWING IS PREPARED FOR THE PROPOSAL PURPOSE ONLY AND CAN BE CHANGED ACCORDINGLY IN THE DETAIL DESIGN PHASE.
- NOTES :
- THIS P&ID IS FOR GROUP#10, CONFIGURATION IS IDENTICAL FOR GROUP#20.
 - DRAIN POT AND WARM-UP LINE SHALL BE LOCATED AS CLOSE AS POSSIBLE FROM STEAM TURBINE MAIN STOP & CONTROL VALVE.
 - DRAIN VALVES SHALL BE LOCATED AS CLOSE AS POSSIBLE FROM DRAIN POT.
- * DENOTES EQUIPMENT WILL BE SUPPLIED BY HRSG.
** DENOTES EQUIPMENT WILL BE SUPPLIED BY ST SUPPLIER.

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REV.	DATE	DESCRIPTION	PREPARED	REVIEWED	APPROVED
B	01.OCT.2025	FOR INTERNAL USE			
A	19.SEP.2025	FOR INTERNAL USE			
EMPLOYER					

TENDERER
SAMSUNG **SAMSUNG C&T**
Engineering & Construction Group

PROJECT NAME
AL NOUF1 PROJECT

DRAWING TITLE
P&ID FOR CRH STEAM SYSTEM GROUP 10

PROJECT DWG. NO.	PROJECT NO.	REF. DRAWING NO.	SHEET	REV.
D00P-10LBC40-M05-0001	D00P		1/1	A
			NONE	